

Question Paper

Exam Date & Time: 27-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH EXAMINATIONS - NOVEMBER 2025

Machine Learning and Applications [CSS_5116]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze how reward function design affects the convergence rate in reinforcement learning. (3)

A) Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Apply the concepts of instance-based and model-based learning to illustrate their differences using a practical example. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) A researcher examines how study hours affect student exam scores. Apply a linear regression model to fit, interpret the slope, and predict the score for 7 hours of study. (5)

Student	X (Study Hours)	Y (Exam Score)
1	1	25
2	2	45
3	3	50
4	4	65
5	5	70

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2) Examine how decision trees handle categorical and continuous features. (5)

A) Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Construct a graphical representation showing how changes in the C parameter alter the SVM decision boundary. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Analyze the difference in performance between hard-margin and soft-margin SVM. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3) Categorize the effects of L1 and L2 regularization based on their impact on feature weights and model complexity. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Build a cross-validation framework to demonstrate how it enhances the reliability of training and testing results. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Identify the features of Bayesian models that make them suitable for handling uncertain or incomplete data. (2)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Apply Bayesian inference techniques to design a real-time recommendation system that adapts to user preferences dynamically. (3)
- A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Construct a comparative analysis showing how regularized regression outperforms ordinary least squares in noisy datasets. (2)
- Course Outcome: "CO4" / Cognitive Level: "Application"
- C) Compare the computational efficiency of Ridge and Lasso regression. (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 5) Analyze the role of activation functions in introducing nonlinearity to neural networks. (5)
- A) Course Outcome: "CO5" / Cognitive Level: "Analysis"
- B) Apply Principal Component Analysis (PCA) to illustrate its role in reducing image dimensions for compression. (2)
- Course Outcome: "CO5" / Cognitive Level: "Application"
- C) Develop a perceptron model that can perform logical AND and OR operations. (3)
- Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5116:Machine Learning and Applications

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	10
3	3	CO 3	20
4	4	CO 4	5
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5116:Machine Learning and Applications

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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	9	24(48%)
4	Analysis	6	26(52%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)